BRIDGING NEURAL AND BEHAVIORAL DATA WITH XAI IN PREDICTIVE NEUROMARKETING

25-26J-145-DS

Project Proposal Report

Senanayaka I H D M

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Senanayaka I H D M	IT22120984	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date

....31.08.2025.....

Mr. Samadhi Rathnayake

ABSTRACT

Consumer decisions are affected by both conscious behaviors and subconscious neural processes. Although predictive models in neuromarketing can associate these signals with choices, their internal mechanisms frequently remain as “black box.” This lack of transparency limits trust and raises ethical concerns about bias and manipulation.

This research suggests an ethically transparent XAI framework to bridge this gap. The framework integrates behavioral data and neural signals to predict consumer choices while also providing clear explanations of how decisions are made. Counterfactual explanations offer “what-if” scenarios to show how slight changes in behavior or neural responses could alter outcomes. Ethical scoreboards are introduced to quantify trust, bias, and manipulation risks, giving users a way to assess the fairness and reliability of predictions. In addition, dynamic visualizations are designed to make these explanations accessible for non-technical users such as marketers.

The expected outcome is a transparent and ethical decision support system that makes predictive models both understandable and trustworthy. By explanations in psychological insights, this work aims to increase user confidence while encouraging responsible use of AI in consumer research. The findings are expected to support to more ethical applications of neuromarketing and to the broader endeavour of developing trustworthy AI systems.

1. INTRODUCTION

Understanding consumer decision-making requires analyzing both behavioral patterns and neural responses. Recent findings in neuromarketing have shown that predictive models can uncover hidden insights from EEG signals and behavioral patterns, but these models often remain “black box”. Their lack of transparency limits trust and raises ethical concerns, especially in marketing environments when the distinction between bias and manipulation may be subtle. To address this, XAI techniques are being used increasingly to produce interpretable, trustworthy, and ethically transparent systems.

1.1. Background and Literature Survey

Explainability in AI has significantly developed as a critical area of research to enhance the transparency of black-box predictive models. Techniques like SHAP and LIME, which emphasise feature importance, but offer little contextual meaning for end users, were the main focus of early XAI efforts [1], [2]. Predictive models in neuromarketing are progressively incorporating behavioural and EEG data, however it is still difficult to understand such multimodal, time-series inputs [3], [4].

Recent efforts have extended explainability toward time-series neural data. Research have suggested gradient-based and saliency-based attribution techniques specifically designed for EEG, promising in identifying the moments when brain activity influences decision outcomes [5]. These approaches, mostly stay too much technical and are unable to provide insights into consumer psychology. To make model outputs meaningful for marketers and decision-makers, parallel efforts in the behavioural sciences demonstrate the significance of firmly establishing explanations in cognitive theories, such as attention bias and manipulation risk [6].

Counterfactual explanations have gained traction as a way to generate actionable insights by exploring “what-if” scenarios [7]. In marketing applications, counterfactual reasoning offers a potent way to comprehend how minor changes in consumer behaviour or brain signals could change decisions. Nevertheless, counterfactuals have not been widely used in studies on multimodal consumer data.

Furthermore, ethical concerns are increasingly being addressed in XAI literature, particularly in domains like healthcare and finance [8], [9]. However, there aren't many frameworks that measure ethical aspects including bias, manipulation risk, and trust in neuromarketing settings. Visualisation is also essential; whereas dynamic and user-centred designs could significantly increase adoption in real-world marketing, they are rarely made for non-technical people [10].

1.2. Research Gap

Although progress has been made in explainability for neural and behavioral data, three major gaps remain. First, explanations are not tied to actual consumer understanding since most approaches do not combine behavioural psychology with neural time-series attribution. Second, although counterfactuals provide useful information, their use in neuromarketing applications involving multimodal data has not been fully investigated. Lastly, despite growing worries about the responsible use of AI in influencing consumer decisions, ethical transparency metrics like manipulation risk indicators and trust indices are essentially non-existent in this sector.

1.3. Research Problem

Predictive models that combine neural and behavioral data can effectively forecast consumer decisions, but they remain "black box," psychologically detached, and ethically unverified. Their credibility and practical use in marketing are restricted by their lack of interpretability and ethical transparency, which makes an explainable and ethically sound framework imperative.

2. OBJECTIVES

2.1. Main Objective

To develop an ethically transparent XAI framework that explains consumer decision-making by integrating neural and behavioral data, providing actionable insights while assessing trust, bias, and manipulation risk.

2.2. Specific Objectives

- To design and implement a dual-layer XAI framework that combines behavioral psychology principles with time-series neural data, to enable a deeper understanding of consumer decisions.
- To implement counterfactual explanations to offer actionable insights.
- To create ethical scoreboards that quantify trust, bias, and manipulation risk by providing measurable indicators of ethical decision support.
- To develop dynamic visualizations that help to communicate model decisions for diverse stakeholders.

3. METHODOLOGY

3.1. System Overview

The proposed research introduces an Ethically Transparent XAI Framework to explain consumer decision-making using integrated neural and behavioral data. Data collection (EEG signals and behavioural reactions) is the first step in the system pipeline. Next comes consumer decision prediction modelling. The XAI framework is designed to offer ethical, actionable, and interpretable explanations on top of this predictive layer.

The framework will combine four modules:

- Dual Layer explainability – Time series neural data + Behavioural Psychology Grounding
- Counterfactual Explanations – Actionable what-if scenarios
- Ethical Scoreboards – Trust, Bias, and Manipulation risk indices
- Dynamic Visualization – Interactive dashboards for marketers and researchers

3.2. System Diagram

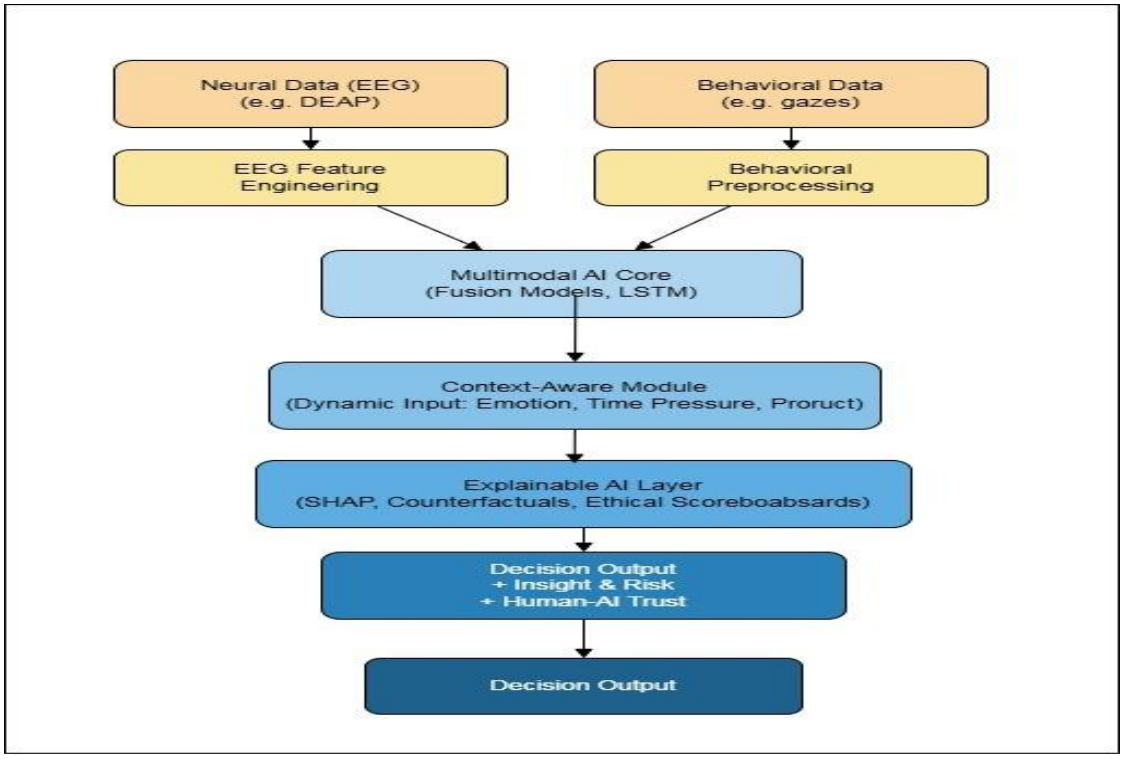

Figure 3.1-System Diagram

3.3. Tasks and Sub-Tasks

Task 1: XAI framework design and development

Sub-task 1.1: Implement Dual Layer Explainability

- Neural Data – Apply time-series feature attribution to highlight which EEG signals and periods affect to decision outcomes.
- Behavioral Data – Map model outputs to established psychological decision factors to explanations more relatable to consumer behavior.

Sub-task 1.2: Generate Counterfactual Explanations

- Create actionable "what-if" scenarios to investigate how minor adjustments to behavioural or neurological signals could affect the choices made by consumers.
- Ensure counterfactuals remain realistic, ethical, and free from manipulative insights.

Sub-task 1.3: Build Ethical Scoreboards

- Design metrics to quantify ethical transparency including Trust Index, Bias Detection, Manipulation Risk Assessment.

Sub-task 1.4: Create Visualization Layer

- Develop interactive dashboards to communicate explanations clearly.
- Tailor visualizations for different stakeholders like marketers, consumers, and researchers with varying levels of technical expertise.

Task 2: Evaluation and Validation

- Technical Validation – Assess accuracy of the explanations compared to predictive model outputs.
- Human-Centered Validation – Conduct usability testing with small groups to measure interpretability and trust.
- Ethical Validation - Evaluate fairness and ethical transparency through scoreboards.

Task 3: Documentation and Dissemination

- Document framework design, implementation process, and experimental results.
- Prepare outputs for academic dissemination, including journal and conference paper drafts and demonstration dashboards.

3.4. Tools and Technologies

Category	Tools
Data Gathering and Preprocessing	EEG Datasets, Pandas, NumPy
Modeling	Scikit-learn, TensorFlow, PyTorch
Explainability Methods	SHAP, LIME, Integrated Gradients, Captum (PyTorch)
Counterfactuals	Dice (Python), Custom scenario modeling
Visualization	Python (Plotly, Matplotlib), Power BI, Tableau
Ethical Assessment	Fairlearn, AIF360 (bias and fairness libraries)

Table 3.2-Tools and Technologies

3.5. Data Requirements

- Neural Data – EEG datasets that include emotional and cognitive responses.
- Behavioral Data – Decision-making records such as consumer choices, reactions, ratings, or preferences.
- Integrated Data - Behavioral labels mapped to EEG time windows to create multimodal features.

Data will be drawn from public freely available neuromarketing datasets and may be complemented with a small-scale survey or experiment for contextual validation like simple product choice tasks.

3.6. Gantt Chart

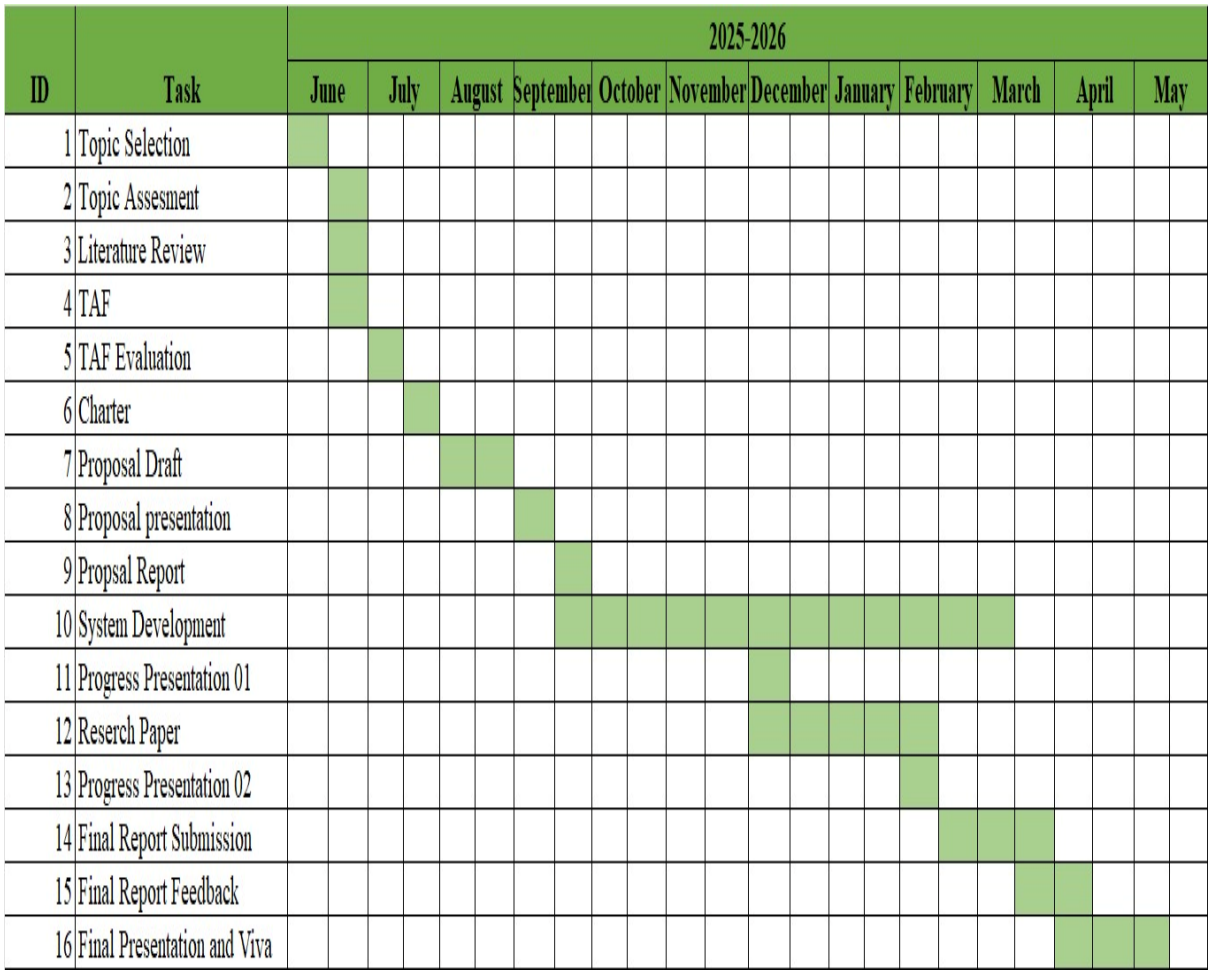

Figure 3.2-Gantt Chart

3.7. Work Breakdown Structure (WBS)

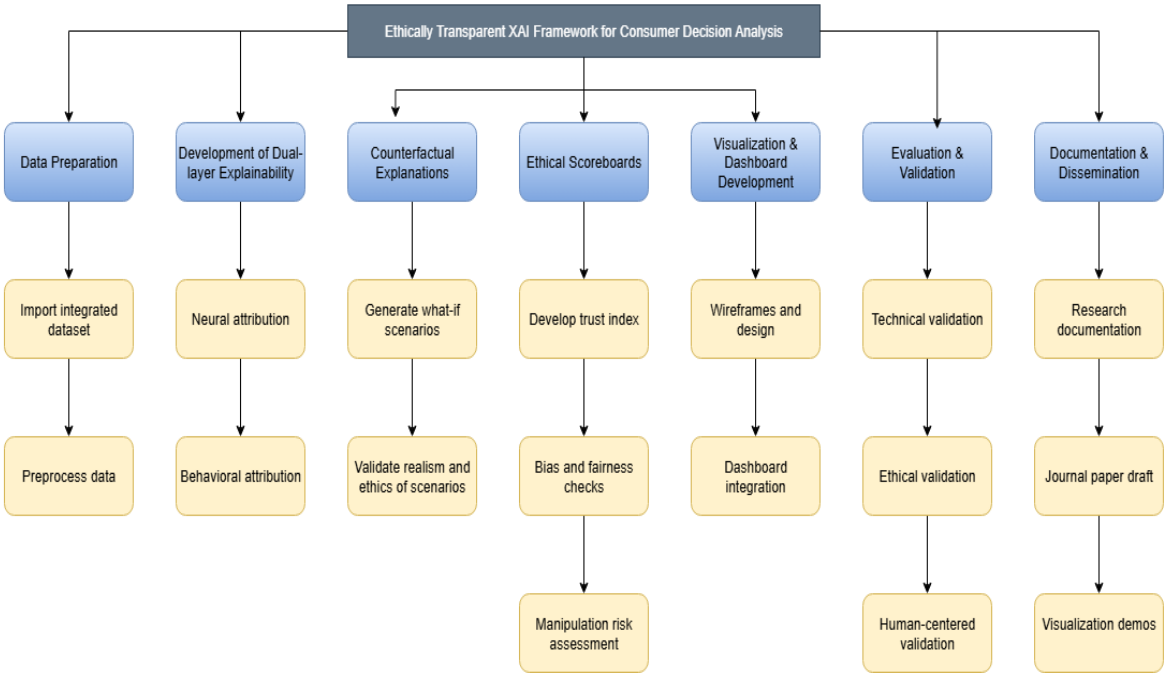

Figure 3.3-WBS

3.8. Real – World Applications

- Help marketers to design marketing campaigns effectively as well as ethically.
- Assists regulators in monitoring manipulation risk in advertising.
- Provides consumers with greater transparency in how their decisions are influenced.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

- Integrate outputs from predictive models trained on neural (EEG) and behavioral data.
- Implement Dual Layer Explainability with Neural time-series attribution and Behavioral attribution.
- Provide counterfactual explanations that suggest realistic “what-if” scenarios for consumer decision changes.
- Create ethical scoreboards quantifying trust, bias, and manipulation risk.
- Visualize explanations dynamically through interactive dashboards tailored for different stakeholders.

4.2. Non-Functional Requirements

- Performance: Generate explanations within seconds.
- Scalability: Handle large datasets like thousands of EEG recordings + behavioral samples.
- Usability: Dashboards should be intuitive for non-technical users like marketers and consumers.
- Reliability: Consistent accuracy of explanations across multiple runs.
- Security and Privacy: Protect sensitive neural and behavioral data, anonymize before analysis.
- Ethical Transparency: Ensure explanations avoid manipulative interpretations.

4.3. User Requirements

- Researchers: Need in-depth explanations and detailed visualizations for validation.
- Marketers: Require simplified, actionable insights.
- Consumers: Need transparency in how their decisions are being influenced.
- Regulators: Require ethical scoreboards to monitor manipulation risk.

4.4. System Requirements

Hardware	• A computer with at least 16 GB RAM, GPU support (NVIDIA) for deep learning. • Storage capacity of 1 TB.
Software	• Python 3.x • ML/DL frameworks (TensorFlow, PyTorch) • Explainability libraries (SHAP, LIME, Captum, DiCE) • Visualization tools (Python Libraries, Power BI, Tableau) • Fairness libraries (AIF360, Fairlearn)

Table 4.1 -System Requirements

4.5. Use Cases (Tentative)

	Use Case	Actors	Description	Outcome
Use Case 01	Generating Explanations	Marketer Researcher	User inputs a consumer decision prediction, and framework generates dual-layer explanation.	User receives feature attributions (neural + behavioral) with clear visualizations.
Use Case 02	Counterfactual Exploration	Marketer Researcher	User requests 'what-if' scenarios for consumer decisions.	The system shows how small changes in neural or behavioral inputs change predictions.
Use Case 03	Ethical Evaluation	Regulator Consumer	User requests an ethical risk analysis of predictions.	Ethical scoreboard displays trust, bias, and manipulation indices.

Table 4.2 -Use Cases

4.6. Test Cases (Tentative)

4.6.1. Technical Test Cases

Test Case ID	Test Case	Purpose	Method	Expected Outcome
TC1	Neural Attribution Validation	Test if framework highlights decisions relevant to EEG signals.	Compare highlighted EEG segments with established cognitive neuroscience markers.	High overlap with known neural decision-making patterns.
TC2	Behavioral Attribution Validation	Test if explanations align with consumer psychology.	Map top features to cognitive biases and cross-check with behavioral literature.	Behavioral explanations reflect psychological theories.
TC3	Counterfactual Validity	Ensure 'what-if' scenarios are realistic.	Generate counterfactuals, verify with domain experts.	Counterfactuals are realistic, ethical, and non-manipulative.

Table 4.3 -Technical Test cases

4.6.2. Ethical Test Cases

Test Case ID	Test Case	Purpose	Method	Expected Outcome
TC4	Trust Index	Assess if explanations increase trust.	Measure user confidence before vs. after explanation.	Higher post-explanation trust scores.
TC5	Bias Detection	Detect unfair influence across demographic groups.	Run model on subsets.	Ethical scoreboard flags disparities.
TC6	Manipulation Risk	Check if explanations could be misused.	Independent reviewers evaluate explanations for manipulative content.	Explanations flagged as low risk.

Table 4.4 -Ethical Test cases

4.6.3. Usability and Human-Centered Test Cases

Test Case ID	Test Case	Purpose	Method	Expected Outcome
TC7	Dashboard Usability	Test clarity and accessibility of visualizations.	Conduct user study with consumers and marketers using dashboards.	80%+ users find explanations clear and easy to interpret.
TC8	Stakeholder Relevance	Test customization for different audiences.	Collect feedback via surveys.	Explanations meet the needs of each stakeholder group.

Table 4.5 -Usability Test cases

4.7. Wireframes (Tentative)

The main elements of the suggested Explainable AI framework for consumer decision analysis are depicted in the wireframes. In addition to behavioural and brain (EEG-based) attributions, the **Dashboard (Wireframe 1)** offers prediction outputs with dual-layer explanations and ethical score indicators. On the **Counterfactual Explanations (Wireframe 2)** page, ethical risks like bias and manipulation are highlighted along with practical "what-if" examples. Lastly, users can examine trust and fairness indicators, evaluate merged neural–behavioural timelines, and export results for additional analysis on the **Visualisation Report (Wireframe 3)** page. These wireframes guarantee interpretability, ethical transparency, and user accessibility by acting as a conceptual roadmap for the system's interaction flow.

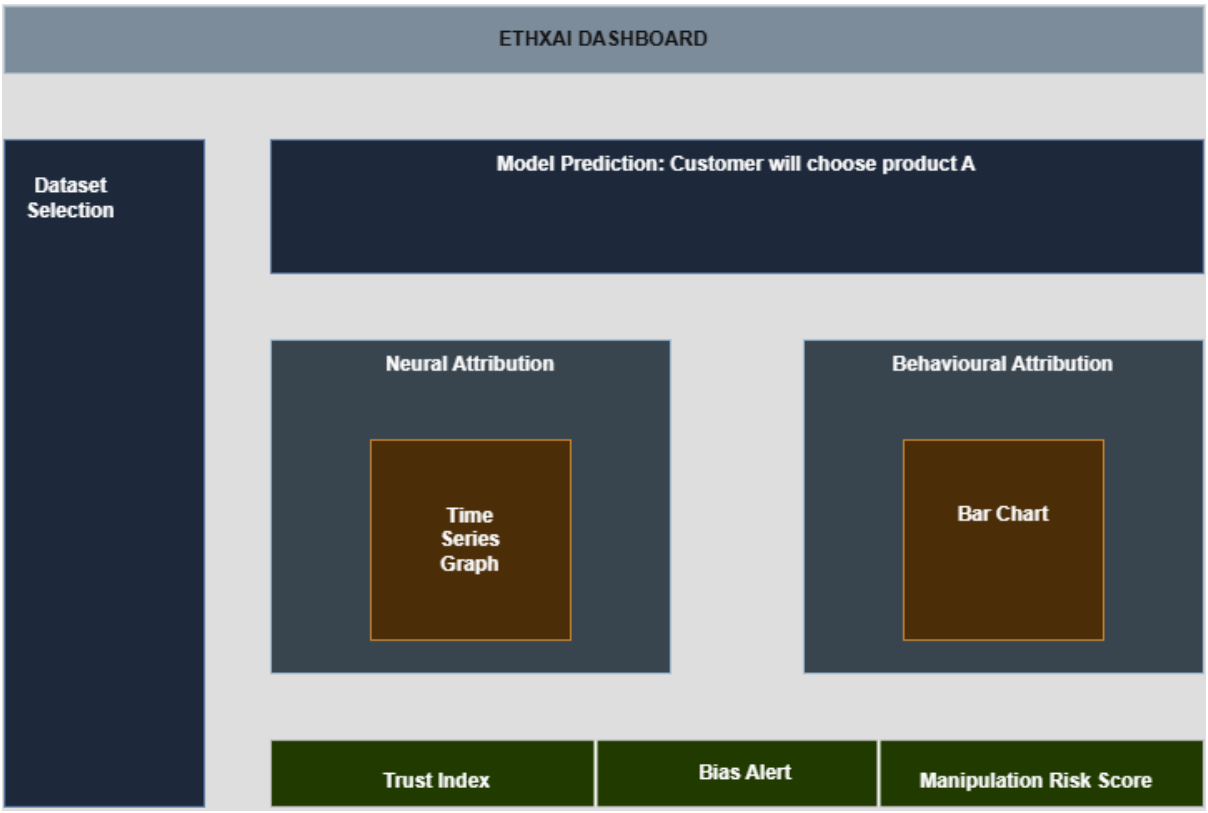

Figure 4.1 – Wireframe 01 (Dashboard)

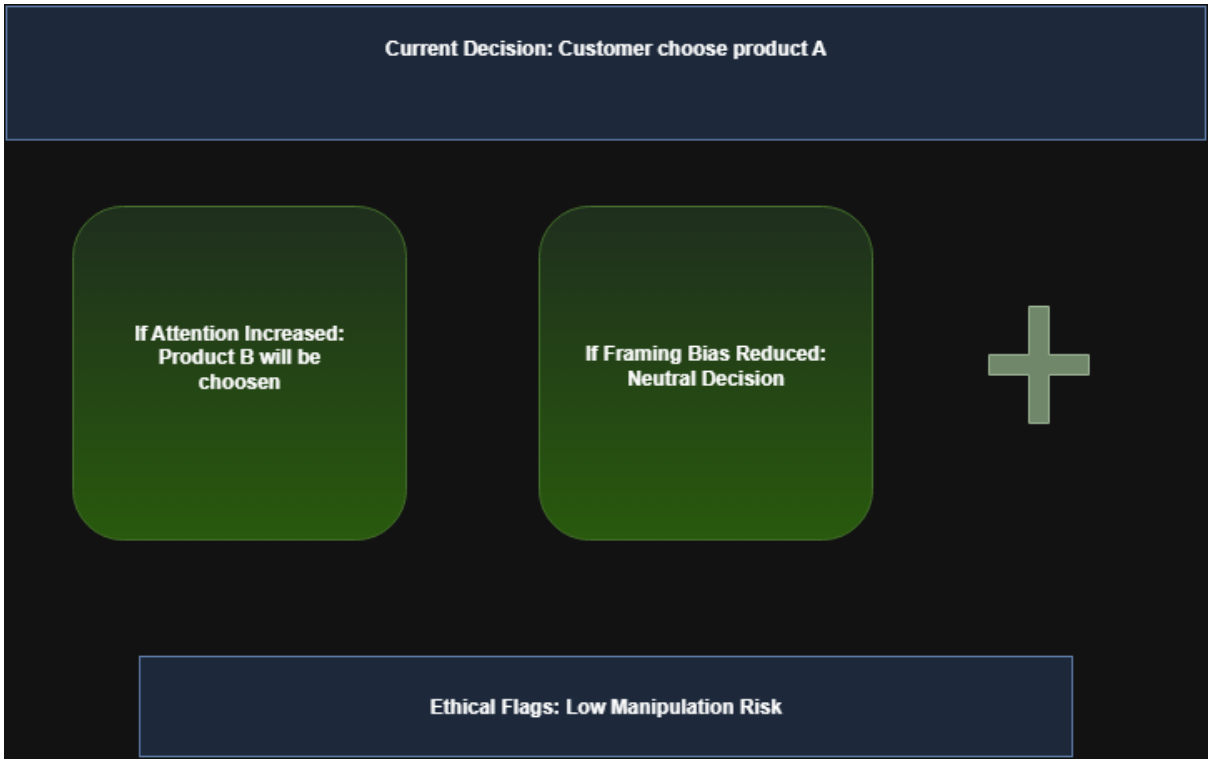

Figure 4.2 – Wireframe 02 Counterfactual Explanations)

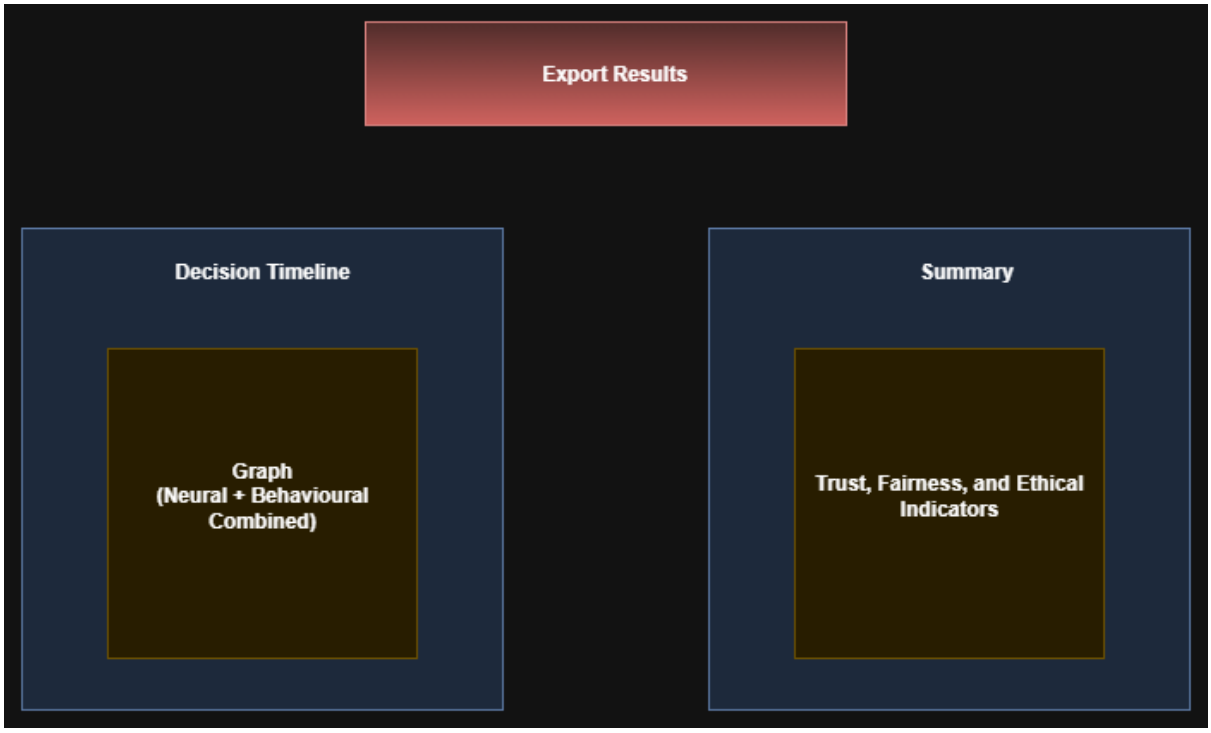

Figure 4.3 – Wireframe 03 (Visualization Report)

5. DESCRIPTION OF PERSONAL AND FACILITIES

5.1. Personal Contribution

With an emphasis on ethical transparency in consumer decision analysis, I will be in charge of creating the Explainable AI (XAI) framework. Producing ethical scoreboards, applying counterfactual explanations, establishing the dual layer explainability model, and producing interactive visualization dashboards are all part of my job. In addition, I will oversee the technical, ethical, and usability testing validation process.

This effort is technically supported by my expertise in data science, machine learning, and model evaluation. To help with the implementation, I'm also getting ready to increase my knowledge of explainable AI through short courses and self-study such as XAI frameworks and fairness tools.

5.2. Facilities and Resources

- Supervisory Support:
  1. Supervisor - Mr. Samadhi Rathnayake
  2. Co-Supervisor – Ms. Samanthi Siriwardene
  3. External Supervisor – Dr. Kamal Gunarathne
- Software Tools: Open-source libraries including TensorFlow, PyTorch, SHAP, LIME, Captum, DiCE, AIF360, Fairlearn, and visualization frameworks.
- Data Resources: Publicly available EEG and behavioral datasets for consumer decision-making scenarios.
- University Facilities: University-provided resources such as GitHub Enterprise, cloud storage, access to the university's digital library, research databases, and journal subscriptions for literature review.

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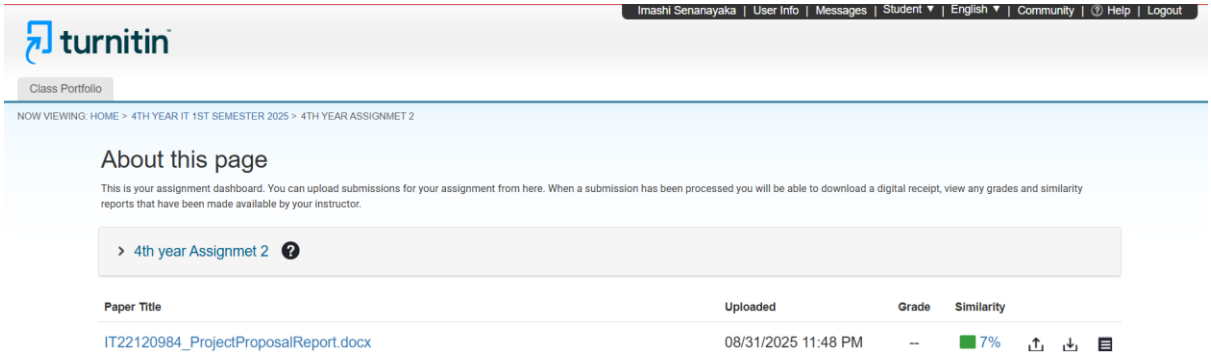
APPENDICES

Appendix A*: Plagiarism*